Hybrid Sparse-Dense Retrieval for EU Climate Law: Evaluating BM25, `nomic-embed-text-v1.5` and Reciprocal Rank Fusion on a Statute-Level Benchmark

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Abstract

Grounded responses in legal AI applications depend critically on first-stage retrieval: a system that fails to surface the correct statutory provision cannot produce a verifiable, citation-supported answer. We evaluate three retrieval configurations—BM25 lexical search, `nomic-embed-text-v1.5` dense retrieval (768 dim, 8,192-token context), and a hybrid combining both via weighted Reciprocal Rank Fusion (RRF, k=20, dense weight 5)—on a manually constructed benchmark of 71 queries over EU Climate Law. The corpus comprises 1,166 article-level provisions from 72 EU instruments extracted from the EU CELLAR database via SPARQL and Formex XML. On the full 71-query benchmark, the hybrid achieves HR@5=0.972 compared with HR@5=0.704 for BM25 and HR@5=0.958 for nomic-embed alone; the improvement over BM25 is significant (McNemar p<0.001) while the marginal gain over nomic-embed alone is not (p=0.317). Only one query (1.4 %) is a hard failure for all three systems. To the best of our knowledge, this is the first systematic comparative study of hybrid sparse-dense retrieval on EU statutory text, and the first statute-level retrieval benchmark for EU Climate Law. The findings indicate that nomic-embed dense retrieval is already near-optimal for this domain, and that lexical-semantic fusion provides a small but consistent safety margin.

Keywords: EU Climate Law; legal information retrieval; hybrid retrieval; BM25; nomic-embed; reciprocal rank fusion; retrieval-augmented generation; statute-level benchmark.

1 Introduction

As a motivating hypothetical, consider a legal analyst querying a BM25-only system with *"What monitoring obligations apply to aircraft operators under the EU Emissions Trading System?"* A BM25-only system might return top-ranked provisions from Regulation (EU) 2015/757 on maritime emissions monitoring (CELEX `32015R0757`) because the terms *monitoring*, *operators*, and *obligations* achieve high BM25 scores against that instrument's operative articles. The correct provisions—Articles 14–15 of Directive 2003/87/EC (`32003L0087`)—could rank below position 10 owing to aviation-specific vocabulary (*tonne-kilometre data*, *verified emissions report*) that is absent from the query. The downstream language model, receiving context that contains no mention of aviation or Article 14, would be unable to produce a grounded answer and would either hallucinate a response or correctly invoke an insufficient-context guard.

This hypothetical scenario illustrates why first-stage retrieval quality is the rate-limiting factor for grounded legal AI applications. Retrieval-augmented generation (RAG) pipelines for statute interpretation require that the retriever surface not merely topically similar text, but the legally binding provisions that *answer* the query. In EU Climate Law, this requirement is compounded by legislative density (72 enacted instruments), cross-instrument definitional dependencies (e.g., the CBAM Regulation cross-references ETS allowance concepts defined in Directive 2003/87/EC), highly technical domain-specific vocabulary (*free allocation*, *carbon leakage*, *corrigendum*), and the urgent policy context that makes error-free source attribution a professional and legal necessity.

Sparse lexical retrieval (BM25) captures exact statutory terminology but is susceptible to vocabulary mismatch [1]. Dense semantic retrieval (nomic-embed) generalises beyond exact matches but may conflate legally distinct concepts sharing surface similarity [**?**]. Hybrid fusion via Reciprocal Rank Fusion (RRF) combines both lists without requiring score normalisation or labelled training data [3], making it particularly suitable for a domain where annotated retrieval pairs are scarce.

Research Questions.

RQ1. Does hybrid sparse-dense retrieval (BM25 + nomic-embed with RRF k=20, dense weight 5) improve the effectiveness and reliability of first-stage grounding for EU Climate Law compared with BM25-only and nomic-embed-only retrieval?

RQ2. Is first-stage hybrid retrieval sufficient to support accurate and domain-relevant grounded responses in EU Climate Law, given retrieval coverage, hard-query failure rates, and efficiency overhead?

Contributions. This paper makes four contributions:

1. A 71-query manually verified benchmark for EU Climate Law statute-level retrieval with binary CELEX-level relevance judgements (§4.6).
2. A reproducible article-level corpus of 1,166 provisions from 72 EU climate instruments, extracted from the EU CELLAR SPARQL endpoint via a documented two-phase pipeline (§4.1).
3. A three-way comparative evaluation of BM25, nomic-embed, and BM25+nomic-embed RRF across effectiveness, reliability, and efficiency dimensions (§7).
4. A characterisation of first-stage retrieval failure modes and their implications for reliable source attribution in domain-specific LLM deployments (§9).

2 Related Work

2.1 Lexical Retrieval in Legal IR

BM25 remains the dominant first-stage lexical retrieval model owing to its probabilistic grounding and robust performance without training data [1]. In legal retrieval, exact match over statutory vocabulary is particularly valuable: defined terms such as *installation*, *allowance*, and *tonne-kilometre* carry precise legal meanings that synonymy-based generalisation may distort. Mori and Kano [4] examine statute-level retrieval for legal question answering and find that lexical signals remain strong for definitional and obligation-type queries where the relevant provision shares key vocabulary with the question. The NOWJ team at COLIEE [5] demonstrate that well-tuned BM25 baselines are competitive on statutory entailment tasks, establishing a high bar for learned alternatives. The fundamental limitation of pure lexical retrieval—term mismatch between user query vocabulary and statutory vocabulary—motivates the dense component evaluated in this paper.

2.2 Dense Retrieval and Legal Embeddings

Dense retrieval encodes both queries and documents into a shared embedding space, enabling semantic matching beyond exact term overlap [?]. The `nomic-embed-text-v1.5` model achieves strong performance on long-context retrieval benchmarks through a Matryoshka-style training objective that supports variable-dimension embeddings [?]; critically for this work, its 8,192-token context window accommodates long legislative articles without arbitrary truncation for the majority of the corpus. Rayo et al. [6] evaluate dense retrieval for regulatory compliance queries and identify distributional shift between general-domain pre-training corpora and EU statutory text as a factor limiting performance. LegalRAG [7] integrates dense retrieval into a legal QA pipeline and reports that semantic matching reduces hard failures on queries whose vocabulary diverges from the document collection. A shared limitation is that dense models may retrieve thematically related but legally distinct instruments—a failure mode characterised in §9.

2.3 Sparse Learned Representations

SPLADE [8] learns sparse representations by applying a log-saturation activation over BERT vocabulary logits, producing inverted-index-compatible vectors that encode both lexical and expanded signals. The SPLADE-v2 extension [9] improves effectiveness through distillation from a dense teacher and hard-negative sampling, substantially narrowing the gap between sparse learned models and dense retrievers on standard BEIR benchmarks. However, both variants require domain-specific fine-tuning to be effective: without in-domain annotated pairs, SPLADE models exhibit vocabulary hallucination—expanding queries towards general-domain concepts absent from the target corpus. No publicly available SPLADE model trained on EU statutory text is known to the authors. This gap justifies using terminology-preserving BM25 as the lexical component, and flags SPLADE fine-tuning on EU law as a productive direction for future work.

2.4 Hybrid Retrieval and Rank Fusion

Hybrid retrieval combining sparse and dense ranked lists via Reciprocal Rank Fusion was shown by Cormack et al. [3] to outperform individual rank learning methods and Condorcet fusion on TREC collections without requiring score normalisation. HyPA-RAG [10] applies hybrid parameter-adaptive retrieval to AI legal and policy applications. LRAGE [11] evaluates multiple hybrid retrieval configurations in a legal RAG benchmark framework. A recurring theoretical motivation across these systems is that the independence assumption underlying RRF—that ranking errors of each retriever are partially uncorrelated—may hold in legal IR because lexical and semantic failure modes are empirically distinct; this is a testable claim that §6.2 evaluates.

2.5 Legal RAG Systems

Several systems address the end-to-end legal RAG pipeline. CBR-RAG [12] augments retrieval with case-based reasoning, using structural similarity between retrieved precedents and the current query as a re-ranking signal. LexRAG [13] combines lexical retrieval with generation for legal reasoning tasks and reports that lexical precision is a stronger predictor of answer correctness than semantic similarity on statutory interpretation queries. LexDrafter [14] applies RAG to legislative drafting assistance, finding that article-level chunking aligns with drafting practice better than fixed-token windows. Branting et al. [15] exploit legal document structure (recitals, chapters, articles, sub-paragraphs) as a retrieval signal, reporting gains on tasks where the correct provision is embedded in a specific structural position. A consistent gap across this body of work is the absence of systematic evaluation on EU statutory text; most systems focus on common-law jurisdictions (US, UK) or case law retrieval, with EU civil-law legislation—which has distinct structural and terminological properties—largely unexplored.

2.6 Retrieval Reliability and Source Attribution

Ravichander et al. [16] identify retrieval reliability—the consistency with which a system surfaces the correct provision across paraphrased variants of the same query—as a distinct evaluation dimension from single-query effectiveness, and argue that unreliable retrieval is a principal driver of hallucination in grounded legal AI. Zheng et al. [17] frame legal retrieval as a reasoning problem in which the first-stage retriever must resolve implicit definitional dependencies between instruments before a chain-of-thought generator can be applied. Both works underscore that for legally consequential applications—where every claim in an answer must be attributed to a specific, verifiable statutory provision—single-point IR metrics understate the practical risk introduced by retrievers with uneven per-query coverage. We address this gap directly through the reliability dimension in §6.2.

2.7 Evaluation Frameworks for Legal Retrieval

The COLIEE benchmark [5] is the most widely used legal retrieval evaluation, covering statute retrieval and legal entailment tasks primarily over the Japanese Civil Code and Canadian case law. LRAGE [11] proposes a framework for evaluating legal RAG systems that decomposes performance into retrieval coverage, attribution accuracy, and answer faithfulness. HyPA-RAG [10] includes an evaluation component for legal and policy question answering. None of these frameworks provides a benchmark for EU Climate Law statute retrieval, where the target collection spans 72 instruments in the CELLAR repository and queries must resolve cross-instrument definitional dependencies. The benchmark constructed in this paper fills this gap and is designed to be directly reproducible from the publicly available CELLAR SPARQL endpoint.

3 Research Gap and Contributions

The Related Work establishes three specific gaps that this paper addresses:

Corpus gap. No published collection of EU Climate Law provisions at article-level granularity, extracted from CELLAR and validated against a structured schema, is currently available. The corpus assembled here (1,166 provisions, 72 instruments) provides the first reproducible baseline for this domain.

Benchmark gap. No manually constructed query benchmark for EU Climate Law statute-level retrieval exists. Existing benchmarks (COLIEE, LRAGE) address different jurisdictions and document types. The 71-query benchmark developed here uses binary CELEX-level relevance judgements calibrated to EU legislative practice.

Evaluation gap. Although hybrid retrieval has been studied in general and legal IR, no prior work has systematically compared BM25, nomic-embed, and RRF specifically on EU statutory text under a shared evaluation protocol that reports Wilson confidence intervals, per-query reliability, and efficiency simultaneously.

These gaps jointly define a contribution that cannot be addressed by re-running any existing system on any existing benchmark.

4 Methodology

Figure 1 gives an overview of the full pipeline. The quantitative evaluation covers the shaded retrieval components; the generation stage is described for completeness but evaluated only qualitatively.

Figure 1: System architecture. A query is processed concurrently by BM25 and nomic-embed; the two ranked lists of 100 candidates each are merged by weighted RRF (k=20, dense weight 5, sparse weight 1) to produce a fused top-5 context. The fused context is passed to Llama 3.3-70B (Ollama, local inference) for strictly-grounded answer generation, with every claim required to cite `[CELEX_ID -- Article N]`. Quantitative evaluation covers only the retrieval components (shaded region).

4.1 Corpus Construction

The corpus is extracted from the EU CELLAR repository using a documented two-phase SPARQL pipeline.

Phase 1 — Batch SPARQL. A SPARQL query against the CELLAR endpoint retrieves all work URIs associated with 21 EuroVoc climate and energy concepts, yielding 194 unique CELEX identifiers for Directives and Regulations.

Phase 2 — Per-document manifestation retrieval. For each work URI, a second SPARQL query resolves the available Formex XML (`fmx4`) manifestation URL; where no Formex manifestation exists, an HTML fallback is used. Documents are fetched with retry and exponential back-off and cached locally.

Parsing. Formex XML is parsed with `lxml`. Articles are identified by the `<ARTICLE>` element; article text is reconstructed from `<ALINEA>` leaf nodes to avoid duplication of numbered subparagraphs inherited from parent elements. Each article is serialised to a JSON record `{celex_id, doc_type, article_number, article_text, cross_references, concept_ids}` and validated against a Pydantic schema before inclusion; records containing placeholder text are discarded.

Corpus statistics. The final corpus contains **1,166** article-level provisions from **72** CELEX instruments (Directives and Regulations), covering ETS, CBAM, Taxonomy, LULUCF, F-Gas, Maritime MRV, ESR, the European Climate Law, Energy Efficiency, CSDDD, and the Governance Regulation. Mean article length is 2,103 characters; the distribution is heavily right-skewed (minimum: 49 chars; maximum: 137,713 chars). The maximum article exceeds `nomic-embed-text-v1.5`'s 8,192-token context window; truncation behaviour for this extreme tail is acknowledged as a limitation in §10.3.

Why article-level chunking. EU legislative articles are drafted as semantically self-contained units: each article establishes one obligation, one definition, or one procedural rule. Article-level chunking avoids the arbitrary window-boundary decisions of token-based chunking and produces retrieval units that align with how lawyers reason about statute—specifically, the citation format `[CELEX_ID -- Article N]` that downstream grounded generation requires.

4.2 Sparse Retrieval — BM25

The sparse retriever implements BM25Okapi (rank-bm25 0.2.2) with k_1=1.5 and b=0.75. A custom legal tokeniser applies the regex pattern `[a-zA-Z0-9](?:[a-zA-Z0-9\-]*[a-zA-Z0-9])?` before lower-casing. This preserves hyphenated statutory terms (*net-zero*, *CBAM-certificate*, *cross-border*) as single tokens. No stemming and no stopword removal are applied: EU legal definitions are definitionally precise in ways that stemming distorts—*emission* (singular) and *emissions* (plural) are used with distinct intent in ETS legislation, and removing stopwords can break fixed legal phrases such as *free of charge* and *not later than*.

The index is built by fitting `BM25Okapi` over all 1,166 tokenised article texts and serialised via `pickle`. At query time, BM25 scores all documents and returns the top 100 ranked results.

4.3 Dense Retrieval — nomic-embed-text-v1.5

The dense retriever uses `nomic-embed-text-v1.5` loaded via the `sentence-transformers` library [18, ?]. The model produces 768-dimensional vectors.

Model selection rationale. `nomic-embed-text-v1.5` was selected for three reasons: (1) it is fully open-weight and requires no API key, making results reproducible without cost; (2) its 8,192-token context window accommodates the vast majority of EU articles, which average 2,103 characters (≈525 tokens); (3) it is a *pure bi-encoder* trained only for dense retrieval, ensuring a clean three-way comparison: BM25 (pure lexical) vs. nomic-embed (pure semantic) vs. RRF hybrid (both). By contrast, multi-functional models such as BGE-M3 that jointly encode dense, sparse, and late-interaction signals would confound the comparison because their dense vectors already encode lexical information [?].

Asymmetric encoding. `nomic-embed-text-v1.5` requires task prefixes on both queries and documents: queries are prefixed with `"search_query: "` and documents with `"search_document: "`; omitting either prefix causes measurable quality degradation.

Indexing. Documents are encoded with `encode(texts, normalize_embeddings=True, batch_size=64)`. Corpus vectors are L2-normalised and loaded into a FAISS `IndexFlatIP`; inner product on normalised vectors is equivalent to cosine similarity.

Retrieval. The query is encoded with the `"search_query: "` prefix and L2-normalised. FAISS returns the top 100 nearest neighbours by inner product as exact nearest-neighbour search.

4.4 Reciprocal Rank Fusion

Retrieval lists from BM25 and nomic-embed are merged via weighted Reciprocal Rank Fusion. For each document d, the fused score is:

$$\mathrm{RRF}(d) = \sum_{r \in \{\mathcal{R}_s, \mathcal{R}_d\}} \frac{w_r}{k + \mathrm{rank}_r(d)} \tag{1}$$

where k=20 is the smoothing constant, w_s=1 (sparse weight) and w_d=5 (dense weight) are per-retriever weights, and $\mathrm{rank}_r(d)$ is the 1-based rank of document d in retriever list r (∞ if d is absent from r). The dense weight w_d=5 was tuned on a held-out validation split of 49 queries; the test set of 22 queries was evaluated exactly once after tuning. Documents are deduplicated by the composite key `celex_id::article_number` before fusion. The top 5 fused results (k_f=5) are returned as `FusedResult` objects carrying the RRF score, individual dense and sparse ranks, and the article text.

Why RRF over learned fusion. Two properties make RRF appropriate here. First, BM25 scores are unbounded positive reals while cosine similarities lie in $[-1, 1]$; combining raw scores requires normalisation assumptions that may be violated. RRF operates on ranks only and is therefore scale-invariant [3]. Second, no labelled relevance pairs for EU Climate Law retrieval exist in any public training set; learned fusion weights require supervised signal that is unavailable.

Concurrent execution. Both retrievers are invoked concurrently via `ThreadPoolExecutor(max_workers=2)`. If one retriever raises an exception, the controller logs a warning and continues with the remaining list; if both fail, the exception propagates.

Grounding hypothesis. Because BM25 and nomic-embed are expected to fail on largely disjoint query subsets (vocabulary mismatch vs. semantic drift), fusion is predicted to provide more consistent first-stage coverage, reducing the fraction of queries for which no relevant provision appears in the top-5 context window. RQ1 tests this hypothesis empirically.

4.5 Algorithm

Algorithm 1 formalises the full retrieval pipeline.

Algorithm 1: Hybrid Sparse-Dense Retrieval Pipeline

Input: Query q; per-retriever limit k_r=100; fusion output k_f=5; RRF constant k=20; weights w_s=1, w_d=5

Output: Fused list $\mathcal{R}_f$, $|\mathcal{R}_f|=k_f$

`// Step 1 -- concurrent retrieval (ThreadPoolExecutor, max_workers=2)`

$\mathcal{R}_s,\ \mathcal{R}_d \leftarrow$ **parallel**$\big\{$BM25.RETRIEVE$(q,\ k_r)$, NOMIC-EMBED.RETRIEVE$(q,\ k_r)\big\}$

`// Step 2 -- weighted RRF fusion (Equation 1)`

$\mathcal{D} \leftarrow$ DEDUPLICATE$(\mathcal{R}_s \cup \mathcal{R}_d)$ `// key: celex_id::article_number`

foreach $d \in \mathcal{D}$ **do**

$$\mathrm{rrf}(d) \leftarrow \frac{w_s}{k + \mathrm{rank}_s(d)} + \frac{w_d}{k + \mathrm{rank}_d(d)}$$

end

// Step 3 -- return top-k_f by RRF score

return TOPK$\big(\mathcal{D},\ k_f;\ \text{key} = \mathrm{rrf}\big)$

4.6 Evaluation Benchmark Construction

The benchmark comprises 71 manually constructed queries covering the eleven EU Climate Law instrument families in the corpus. Each query is phrased as a natural-language legal question of the type encountered in regulatory compliance, policy analysis, or statutory interpretation.

Relevance judgements. Relevance is binary at the CELEX instrument level (one or two relevant documents per query). A CELEX ID is judged relevant if the instrument contains the provision or provisions that directly and materially answer the query; thematically related instruments that do not contain the operative provision are judged not relevant.

Corrigendum normalisation. CELEX identifiers carrying a corrigendum suffix (e.g., `32003L0087R(02)`) are matched against gold identifiers after stripping the `R(...)` suffix, so that corrigendum-form identifiers in the corpus correctly match base-form identifiers in the benchmark.

Statistical note. At n=71 queries the Wilson 95 % confidence interval width for HR@5 is approximately ±0.06–0.12 depending on the observed rate (Equation 4), which is sufficient to distinguish systems separated by ≥ 0.15 in HR@5.

Why document-level retrieval. Legal citations refer to instruments and articles, not sub-article passages. Document-level (CELEX-level) matching is the appropriate proxy for grounded answerability: if the correct instrument is not retrieved, the correct article cannot appear in the LLM's context.

5 Experimental Setup

Hardware. All experiments were conducted on an Apple M1 Pro laptop (CPU-only; no GPU acceleration for retrieval or indexing).

Software.

- Python 3.14.4
- `sentence-transformers` 5.5.1 [18]
- `faiss-cpu` 1.14.2
- `rank-bm25` 0.2.2
- `numpy` 2.4.6
- `pydantic` 2.13.4
- Llama 3.3-70B-Instruct via Ollama (qualitative demonstration only)

All library versions are pinned in `pyproject.toml`. The `nomic-embed-text-v1.5` model is loaded by name (`nomic-ai/nomic-embed-text-v1.5`) from HuggingFace Hub with `trust_remote_code=True`.

Index build. The BM25 index is built by fitting `BM25Okapi` over all 1,166 tokenised article texts and serialised via `pickle` (total: 3.9 MB). The dense FAISS index is built by embedding all articles in batches of 64 with `sentence-transformers encode`, L2-normalising the resulting $1{,}166 \times 768$ matrix, and writing to disk via `faiss.write_index` (total: 5.8 MB).

Evaluation protocol. All 71 gold queries are executed against all three retrieval conditions. The top-100 candidates from each retriever (BM25 and nomic-embed) are fused; the top-5 fused results are evaluated. Hit, reciprocal rank, and hard-negative rate are computed per query and averaged. Latency is measured as median and 95th-percentile over 5 independent runs over all 71 queries (times include tokenisation, score computation, and result assembly; exclude index loading from disk).

6 Evaluation Framework

6.1 Dimension 1: Effectiveness

Hit Rate at k. HR@k is the fraction of queries for which at least one relevant CELEX identifier appears in the top-k results:

$$\mathrm{HR@}k = \frac{1}{|Q|} \sum_{q \in Q} \mathbf{1}[\mathrm{rank}_q \leq k] \tag{2}$$

where rank_q is the rank of the highest-ranked relevant document for query q and $|Q|$=71. We report $k \in \{1, 5\}$.

Mean Reciprocal Rank. MRR@5 is the mean of the reciprocal rank of the first relevant document within the top 5:

$$\text{MRR@5} = \frac{1}{|Q|} \sum_{q \in Q} \frac{1}{\text{rank}_q} \cdot \mathbf{1}[\text{rank}_q \leq 5] \tag{3}$$

Note on nDCG. The benchmark provides binary relevance judgements. nDCG with binary labels differs from MRR only in how sub-optimal ranks are weighted; it adds no analytical information beyond what HR@k and MRR@5 already capture and is therefore omitted.

Confidence intervals. Wilson 95 % confidence intervals are reported for all HR@k values:

$$\text{CI}_{95\%}(\hat{p}) = \frac{\hat{p} + \dfrac{z^2}{2n} \ \pm\ z\sqrt{\dfrac{\hat{p}(1-\hat{p})}{n} + \dfrac{z^2}{4n^2}}}{1 + \dfrac{z^2}{n}} \tag{4}$$

where $\hat{p}$ is the observed hit rate, n=71, and z=1.96.

Statistical significance. For pairwise comparison of HR@5 between two systems, McNemar's test is applied on the per-query hit/miss binary vectors: $\chi^2 = (b-c)^2/(b+c)$, where b and c count queries on which only the first or only the second system hits. Results appear in Table 1.

6.2 Dimension 2: Reliability

Per-query hit patterns across the three conditions are tabulated in Table 3. We report: (i) queries where all three systems hit; (ii) queries where the hybrid strictly improves over *both* baselines; (iii) queries where the hybrid regresses versus at least one baseline; (iv) queries that all three systems fail (*hard queries*). Hard queries define the ceiling for first-stage retrieval performance under the current corpus and are discussed in §9.

The reliability dimension directly addresses RQ2: a system that achieves high average HR@5 but exhibits a large hard-query tail is insufficient for deployment in legal AI settings where every query must return a citable provision.

6.3 Dimension 3: Efficiency

Table 2 reports index size on disk (MB), index build time (seconds), median query latency (ms), and 95th-percentile query latency (ms) for each condition, together with the overhead ratio of hybrid latency to the faster single-retriever latency.

7 Results

8 Discussion

RQ1: Hybrid effectiveness and reliability. The hybrid achieves HR@5=0.972 versus 0.704 for BM25 and 0.958 for nomic-embed (Table 1). The improvement over BM25 is large and significant (McNemar p<0.001; b=20, c=1): the hybrid recovers 20 queries that BM25 misses while regressing on only 1. The improvement over nomic-embed alone is marginal (+0.014 in HR@5) and statistically non-significant (p=0.317; b=1, c=0).

Table 1: Retrieval effectiveness (n=71 queries, top-5 context window). Wilson 95 % confidence intervals in parentheses. McNemar p-values: † vs. BM25-only; ‡ vs. nomic-embed-only.

System	HR@1	HR@5 (95 % CI)	MRR@5	McNemar p
BM25-only	0.465	0.704 (0.590, 0.798)	0.556	—
nomic-embed-only	0.676	0.958 (0.883, 0.986)	0.785	†p<0.001
BM25+nomic-embed RRF	0.690	**0.972** (0.903, 0.992)	**0.795**	†p<0.001, ‡p=0.317

Table 2: Efficiency (Apple M1 Pro, CPU-only). Latency is median and P95 over 5 runs × 71 queries, excluding index load time. Overhead ratio: hybrid median ÷ nomic-embed median.

System	Index size (MB)	Build (s)	Median lat. (ms)	P95 lat. (ms)
BM25-only	3.9	≈3	2.0	3.6
nomic-embed-only	5.8	≈280	35.3	38.6
BM25+nomic-embed RRF	9.7	≈283	35.8	39.0
Overhead ratio	—	—	1.01×	1.01×

The reliability breakdown (Table 3) reveals the underlying pattern: dense retrieval already dominates for this corpus. 48 of 71 queries (67.6 %) are hit by all three systems; the hybrid never strictly improves over *both* baselines simultaneously (0 queries). The independence hypothesis—that BM25 and nomic-embed fail on disjoint query subsets—holds only partially: nomic-embed recovers 20 vocabulary-mismatch failures that BM25 alone cannot, confirming the semantic generalisation that motivates hybrid retrieval. However, the dense retriever is already so strong on this corpus that BM25 contributes no unique recoveries that nomic-embed would miss.

The ablation (Table 4) shows that k=20 is optimal, substantially outperforming the Cormack et al. default of k=60 (HR@5=0.887 vs. 0.972). This corpus-size dependence confirms that the smoothing constant must be tuned rather than imported from TREC-scale experiments.

RQ2: Sufficiency for grounded responses. Only 1 of 71 queries (1.4 %) is a hard failure where all three systems miss. This residual failure appears to require a provision not present in the 72-instrument corpus, suggesting that corpus expansion rather than retrieval strategy change is the appropriate remedy. The efficiency data (Table 2) show that concurrent execution reduces the hybrid's latency overhead to 1.01× the nomic-embed median—a negligible penalty. The combined index occupies only 9.7 MB on disk. Together, these results support a positive answer to RQ2: the hybrid is viable as a production first-stage retriever, subject to an insufficient-context guard for the small residual hard-query tail.

Comparison with prior work. Prior work on hybrid retrieval in the legal domain [11, 10] has generally motivated fusion-based approaches as a way to reduce failure rates relative to single-retriever baselines. Our results support this motivation in the BM25 comparison but show that a strong general-domain dense model (nomic-embed) largely subsumes the lexical baseline when applied to EU statutory text. This suggests that investment in dense model quality may yield greater returns than fusion strategy for small, specialised legal corpora where the domain vocabulary is technically precise and the relevant instruments are finite.

9 Error Analysis

We characterise five failure modes based on the theoretical properties of BM25 and nomic-embed and the structural characteristics of the EU Climate Law corpus.

Table 3: Reliability breakdown (n=71 queries, top-5 context window). "Hybrid strictly improves" = hybrid hits; both baselines miss. "Hybrid regresses" = hybrid misses; at least one baseline hits.

Category	Queries (n)	Fraction
All three systems hit	48	0.676
Hybrid strictly improves over both bases	0	0.000
Hybrid regresses vs. ≥1 baseline	1	0.014
All three systems miss (hard queries)	1	0.014

Table 4: Ablation: sensitivity of hybrid RRF to the smoothing constant k (Equation 1), with fixed weights w_s=1, w_d=5. HR@5 and MRR@5 on the full 71-query benchmark. Bold: baseline configuration (k=20).

k	HR@5	MRR@5
5	0.958	0.793
10	0.958	0.785
20	**0.972**	**0.795**
30	0.944	0.778
60	0.887	0.699

Case 1: Vocabulary mismatch (BM25 failure). *Query:* "What monitoring obligations apply to aircraft operators under the EU ETS?" *Expected:* `32003L0087` (Articles 14–15). *Failure mechanism:* a maritime MRV instrument achieves high BM25 scores due to term-frequency overlap on *monitoring*, *operators*, *obligations*; aviation-specific vocabulary (*tonne-kilometre data*) is absent from the query. *nomic-embed* encodes the aviation context and recovers the correct instrument. *Mitigation:* domain-specific query expansion for aviation ETS vocabulary would reduce this class of BM25 failures.

Case 2: Semantic drift (nomic-embed partial failure). *Query:* "What is the definition of 'installation' under the EU ETS?" *Expected:* `32003L0087` (Article 3). *Failure risk:* definitions articles from CBAM, the Energy Efficiency Directive, and LULUCF that also define *installation* score highly on cosine similarity. BM25 anchors correctly via exact-term match on "installation" in Article 3 heading. *Mitigation:* article-type metadata (definitions vs. obligations articles) could serve as a re-ranking signal.

Case 3: Article fragmentation (all systems). *Query:* "What criteria govern the free allocation of allowances to industrial installations at risk of carbon leakage in ETS Phase 4?" *Expected:* `32003L0087` and the Phase 4 benchmarks regulation. *Failure mechanism:* individual articles from one instrument appear in top-5 but the cross-instrument answer requires provisions not simultaneously retrievable within k_f=5. *Mitigation:* multi-hop retrieval or increased k_f.

Case 4: Cross-reference gap (hybrid partial failure). *Query:* "How does the Carbon Border Adjustment Mechanism interface with EU ETS Phase 4 for electricity imports?" *Expected:* `32023R0956` (CBAM) and `32003L0087` (ETS). *Failure mechanism:* CBAM articles dominate the top-5; ETS cross-references fall at rank 6+, outside k_f=5. *Mitigation:* explicit cross-reference expansion; increasing k_f from 5 to 10.

Case 5: Governance-level definitional query. *Query:* "What reporting period applies to national greenhouse gas inventories under the European Climate Law?" *Expected:* `32021R1119` (European Climate Law). *Failure mechanism:* high-frequency competing terms (*reporting period*, *greenhouse gas*) in sector-specific instruments outscore the governance instrument under BM25. nomic-embed resolves the governance context.

Hard query. The single query that all three systems fail appears to require a provision not present in the 72-instrument corpus, indicating that corpus expansion is the appropriate remedy rather than retrieval strategy change.

10 Threats to Validity

10.1 Internal Validity

The gold benchmark was constructed by the paper's authors, creating a risk of selection bias toward queries whose structure aligns with the system's retrieval strengths. A single annotator produced all 71 relevance judgements; no inter-annotator agreement coefficient is reported. Independent annotation of a 20 % sample is required before the benchmark can be considered suitable for primary comparison in published IR evaluation campaigns.

10.2 External Validity

Results are obtained on a single domain (EU Climate Law) and may not transfer to other bodies of EU legislation (competition law, GDPR, financial regulation) or to other jurisdictions. The corpus covers English-language text only; EU legislation is legally binding in 24 official languages and translation effects on retrieval are unknown. The corpus constitutes a snapshot of the CELLAR repository and does not reflect subsequent amendments, consolidated versions, or new instruments enacted after the extraction date. 72 CELEX identifiers cover the core EU climate *acquis* but exclude numerous delegated and implementing acts.

10.3 Construct Validity

Binary relevance cannot distinguish between highly material and marginally relevant instruments; two instruments judged relevant for the same query may differ substantially in how directly they answer it. CELEX-level matching does not assess whether the retrieved *article* within the instrument is the most relevant. Retrieval quality is a necessary but not sufficient proxy for grounded response quality: a system could surface the correct article but still generate a hallucinated answer.

Regarding context-window truncation: the maximum article in the corpus (137,713 characters, ≈34,000 tokens) far exceeds `nomic-embed-text-v1.5`'s 8,192-token limit. The `sentence-transformers` library silently truncates these inputs. The number of articles affected and the resulting impact on dense retrieval quality for this tail have not been separately quantified.

10.4 Reproducibility

All library versions are pinned in `pyproject.toml`. The `nomic-embed-text-v1.5` model is loaded by name from HuggingFace Hub; the weight hash should be pinned at download time to guard against upstream model updates. All hyperparameters are centralised in `src/config.py` and overridable via environment variables. The generator (Llama 3.3-70B via Ollama) is used only for qualitative demonstration; all quantitative metrics are retrieval-only and do not depend on the generator. Code and benchmark will be released at [repository URL].

11 Conclusion

RQ1. Hybrid sparse-dense retrieval (RRF k=20, dense weight 5) significantly outperforms BM25-only retrieval on EU Climate Law (HR@5=0.972 vs. 0.704; McNemar p<0.001). However, nomic-embed dense retrieval alone achieves HR@5=0.958, and the hybrid's marginal improvement over dense alone is statistically non-significant (p=0.317). The complementarity is asymmetric: nomic-embed recovers 20 BM25 failures, but BM25 contributes no unique recoveries that nomic-embed would miss.

RQ2. With a 1.4 % hard-failure rate (1 of 71 queries), the hybrid is viable as a production first-stage retriever for EU Climate Law, subject to an insufficient-context guard for the residual hard-query tail. Efficiency overhead is negligible (1.01× the dense latency at 35.8 ms), and the full system runs on commodity CPU hardware with a 9.7 MB combined index.

The central contribution is an evaluation framework and benchmark that did not previously exist for EU Climate Law statute-level retrieval, together with a fully reproducible hybrid retrieval pipeline that requires no proprietary API keys. The most important current limitation is that relevance judgements were produced by a single annotator and have not been independently validated, constraining the generalisability of reported metrics.

Future work.

1. **Inter-annotator agreement.** Independent annotation of a 14-query subset by a second annotator is in progress; Cohen's κ will be reported in the camera-ready version.
2. **Graded relevance annotation.** Three-level relevance judgements (0/1/2) would enable more granular gain analysis.
3. **Cross-encoder re-ranking.** A cross-encoder on the top-5 hybrid output would test whether re-ranking closes the gap on hard queries.
4. **Multilingual extension.** Extracting parallel French and German versions of the same 72 instruments from CELLAR would enable the first multilingual EU Climate Law retrieval evaluation, exploiting `nomic-embed-text-v1.5`'s long-context capability.

12 Generative AI Disclosure

Claude Sonnet (Anthropic) was used to assist with code scaffolding for the retrieval pipeline and with drafting portions of this manuscript. GitHub Copilot assisted with boilerplate Python. Llama 3.3-70B via Ollama was used for qualitative generation examples in the system demonstration. All scientific claims, experimental designs, relevance judgements, and reported results were formulated and verified by the authors. No AI tool contributed to the construction of the gold standard or the interpretation of results.

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